# ABR BUILD BOOK

*University of California, Irvine*
*Spectre | 2026 – 2027*
*Suspension, Steering, Brakes: Rack and Pinion*

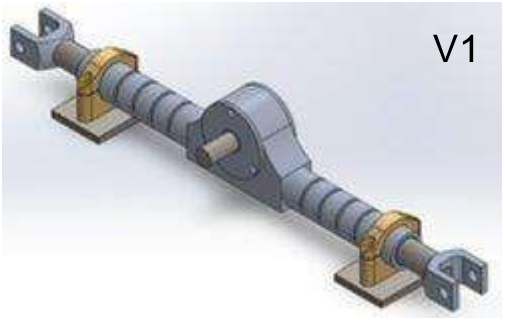
## Design Goals and Constraints

1. Survive calculated loads
2. <190 degree lock to lock
3. Accommodate all mounting points
4. Housing machinable on 3-axis CNC
5. Able to disassemble in <5min
6. No interferences throughout travel

## Trade Study

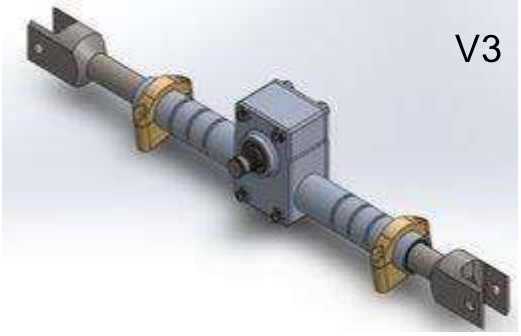
| Name/Link w/ Image | Kaz Rack  | EMPI Buggy Rack  | Custom Rack | NAARCC Rack  |
|---|---|---|---|---|
| Material | Steel gears, Aluminum all else (hollow extensions) | Unknown | Steel gears and extension, aluminum housing | Steel pinion, aluminum rack and housing |
| Ratio/Travel/ Lock-to-Lock | 4.71"/rev, 3.25" travel, 248 deg lock-to-lock | 2.5" travel, 270 deg lock to lock | 6.28"/rev, 3.25" travel, 186 deg lock-to-lock | 4"/rev, 260 deg lock-to-lock |
| Weight/Cost | 3lbs, $850 | 3.33lbs, $215 | 4.4lbs, $194 | 1.5lbs, $650 |
| Notes | 2025 shear extension failure | 2024 bending extension failure | No issues 2026 | Lots of info online |

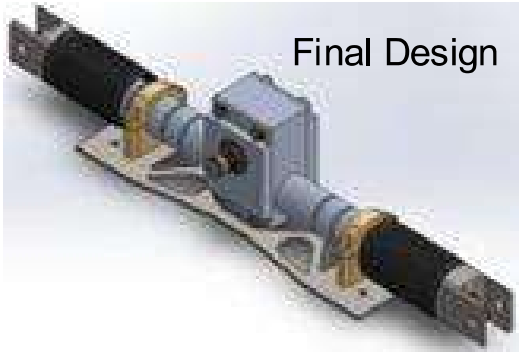
## Design Iterations and Final CAD Model (Designed by Ellie Kirshner)

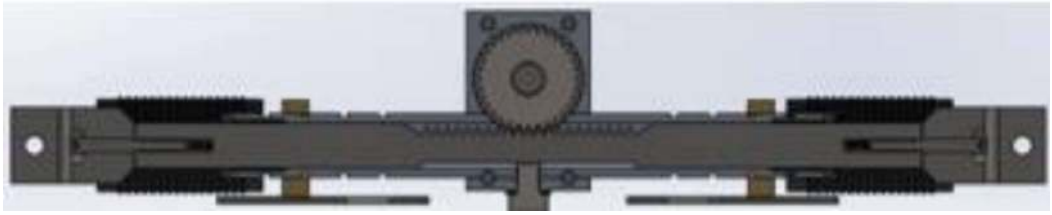








Cutout View of Final Design

# ABR BUILD BOOK

*University of California, Irvine*
*Spectre | 2026 – 2027*
*Suspension, Steering, Brakes: Rack and Pinion*

## Design Decisions

- Major dimensions and mounting match 2025's rack and pinion for interchangeability
- Two part housing to ensure machinability in house (shape of spur gear housing chosen to minimize end mill length needed (absolute max of 2 ⅝" based on Tyler's end mills)
- Gears chosen based on needed ratio for <190 degree lock to lock
- Flat added to bottom of rack gear to restrict rotation with a ½" set screw (machined flat)
- Earlier design had seal on rack gear without the boot, but decided to switch to ensure no dirt gets in, but also lower steering effort (after manufacturing)

## Calculations

**Max Loading Conditions**

| Parameter | Symbol | Value Front | Value Rear | Units | Equation | Notes |
|---|---|---|---|---|---|---|
| Impact Duration | t | 0.25 | 0.25 | sec | | |
| Horizontal Velocity | Vhor | 15.00 | 10.00 | mph | | Horizontal impact (Ex. hitting a log head on) |
| | | 22.00 | 14.67 | ft/s | | |
| Max Horizontal Force | Fh | 1914.59 | 1276.40 | lb | (W/g)*(Vhor/t) | |
| Max Horizontal Force per Tire | Fh,tire | 957.30 | 638.20 | lb | Fh/2 | |

**Tie Rod Calculations**

| Parameter | Symbol | Value | Units | Equation | Notes |
|---|---|---|---|---|---|
| Tie Rod Length | TRL | 16.48 | in | | |
| Front Max Horizontal Force per Tire | FFh,tire | 957.30 | lb | | See Suspension Page (Fh,tire) |
| Torque to Kingpin | TKf | 321.26 | lb*ft | FFh,tire*(SRf/12)/sin(TRA) | Assume impact occurs at max turned tire |
| Force to Tie Rod | FtieR | 1443.85 | lb | TKf/(SALf/12) | |

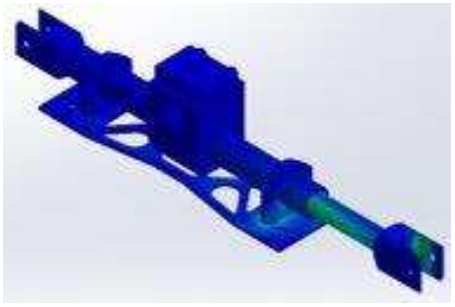
**Rack Loading Conditions**

| Parameter | Symbol | Value | Units | Equation | Notes |
|---|---|---|---|---|---|
| Max Misalignment Angle | MMA | 27.50 | deg | | From max misalignment on tie rod end |
| Max Force to Rack Horizontal | MFRh | 666.69 | lb | FtieR*sin(MMA) | |
| Tie Rod Rack Angle | TRR | 15.60 | deg | | From tie rod angle with rack at ride height (assuming max steering) |
| Max Force to Rack Vertical | MFRv | 388.28 | lb | FtieR*sin(TRR) | |

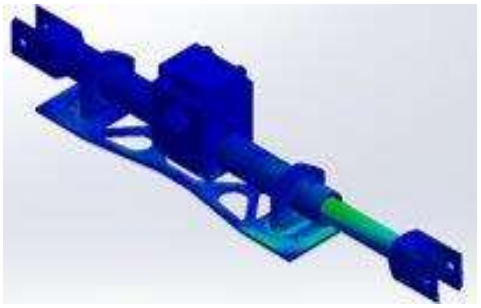
## FEA Results

1. Remove unnecessary components to speed up running FEA (only keep rack gear, housing, pillow block, mounting plate, clevis)
2. Make rack plate fixed on the front and sides where it will be welded to the chassis
3. Apply the calculated force onto the clevis bolts for both the front and vertical impact load cases

**Front Impact**
**80% Yield**



**Vertical Impact**
**68% Yield**



## BOM

| Item | Description | Vendor | Units | Cost/Unit |
|---|---|---|---|---|
| Spur Gear Raw Material | 1144 Carbon Steel Rod, 2 ½" Diameter, ½' Long | McMaster | 1 | $53.19 |
| Rack Gear Raw Material | 1144 Carbon Steel Rod, 1" Diameter, 2' Long | McMaster | 1 | $25.91 |
| Spur Gear Housing Raw Material | Aluminum 6061, 2" x 3 ½" x ½' | McMaster | 1 | $50.62 |
| Rack Gear Housing Raw Material | Aluminum 6061, 1 3/8" OD, ¼" WT, 1' Long | McMaster | 1 | $20.19 |
| Spur Gear Bearing | 5/8" Shaft Diameter Flanged Bearing | McMaster | 2 | $3.96 |
| Spur Gear Seal | 5/8" Shaft Diameter Rotary Seal | McMaster | 2 | $3.33 |
| Rack Gear Bearing | 7/8" Shaft Diameter Sleeve Bearing | McMaster | 2 | $2.95 |
| Assembly Bolt | ¼"-28 x 2" PT Socket Head | McMaster | 1 | $5.49 |
| Assembly Locknut | ¼"-28 Locknut | McMaster | 1 | $5.03 |
| Rack Boot | Universal Rack and Pinion Boot | Amazon | 1 | $12.99 |

**Prototype**



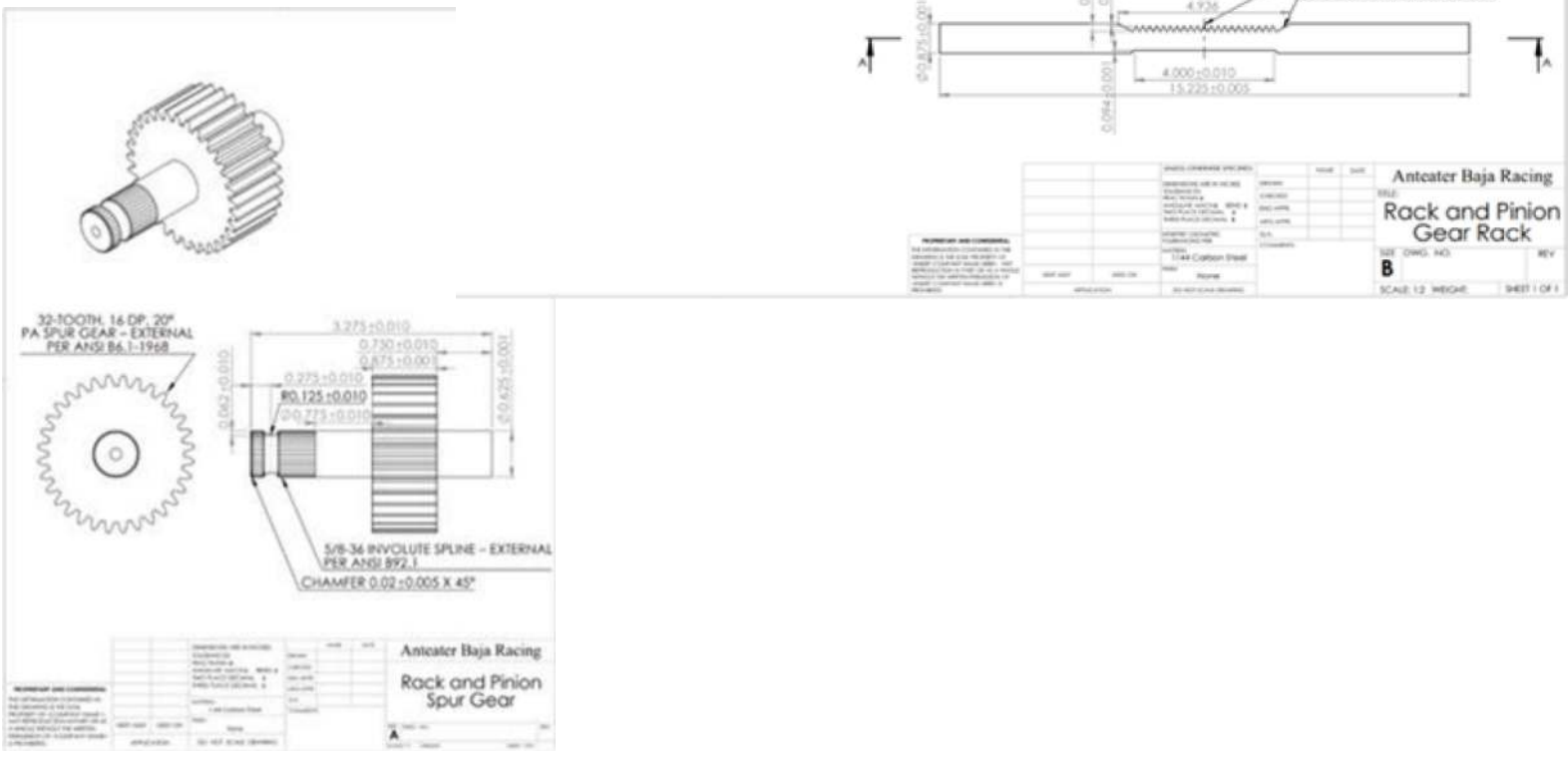
- All components 3D printed on Bambu P1S
  - Housing, gears, bearings printed with PLA
  - Seals printed with TPU
- Confirmed 186 degree lock to lock and fit of all parts



**Manufacturing**

Rack and Spur Gear (Machined by Associated Gears)

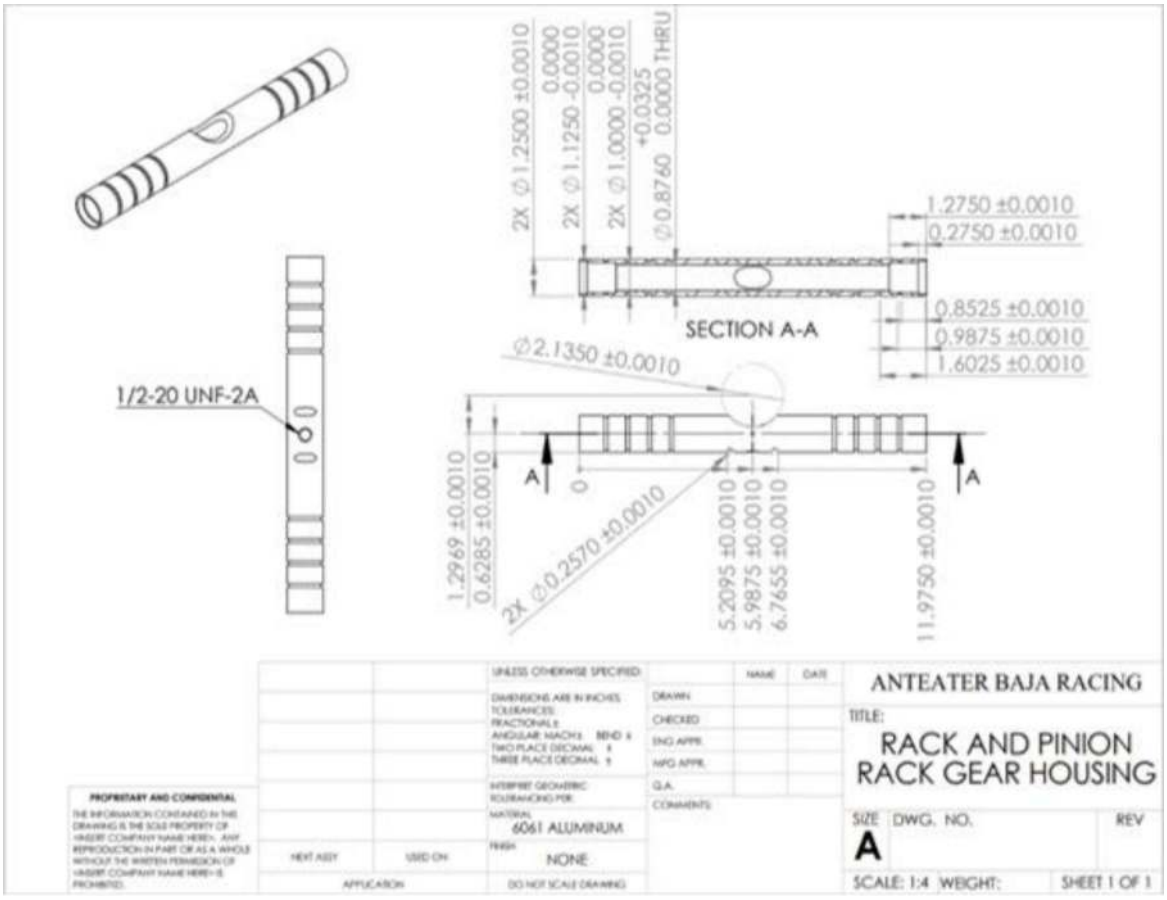
1. Create engineering drawings for outsourcing
    a. Pay attention to tolerancing: used +/- 1 thou for bearings, +/- 5 thou for geometry dimensions, +/- 10 thou for unimportant dimensions
2. Send drawings to machine shop to confirm design feasibility
    a. After discussing with them, learned that they need extra room on either side of rack gear (indicate how much extra room there is for them to cut)
3. Drop off raw material (make sure to order oversized)
4. Pick up completed parts + leftover material (5ish business days)



**Manufacturing**

Rack Gear Housing (Machined by Ellie Kirshner)

1. Put tube into lathe with 6.5” sticking out (use live center)
2. Find x and z zeroes
3. Turn OD down to 1.25” (using “go to” set at 6”)
4. Use groove tool to make pillow block grooves (redo x and z zeroes, 0.075” depth)
    a. Zero the x axis at the first point that you are making the groove
    b. Make sure the z positioning is based on the left edge of the tool and the “go to” value
5. Flip tube in lathe
6. Turn down to 1.25” (record needed “go to” value to intersect the other side)
7. Groove tool again (measure from the intersection point)
8. Face down first side by 0.0125” (only 0.5” sticking out)
9. Face down side side by 0.0125” + extra amount in the “go to” (final length is 11.975”)
10. Drill / ream both ways with closest size to ⅞” to ensure smooth travel (has to be oversized)
11. Turn down ID for bearing housing

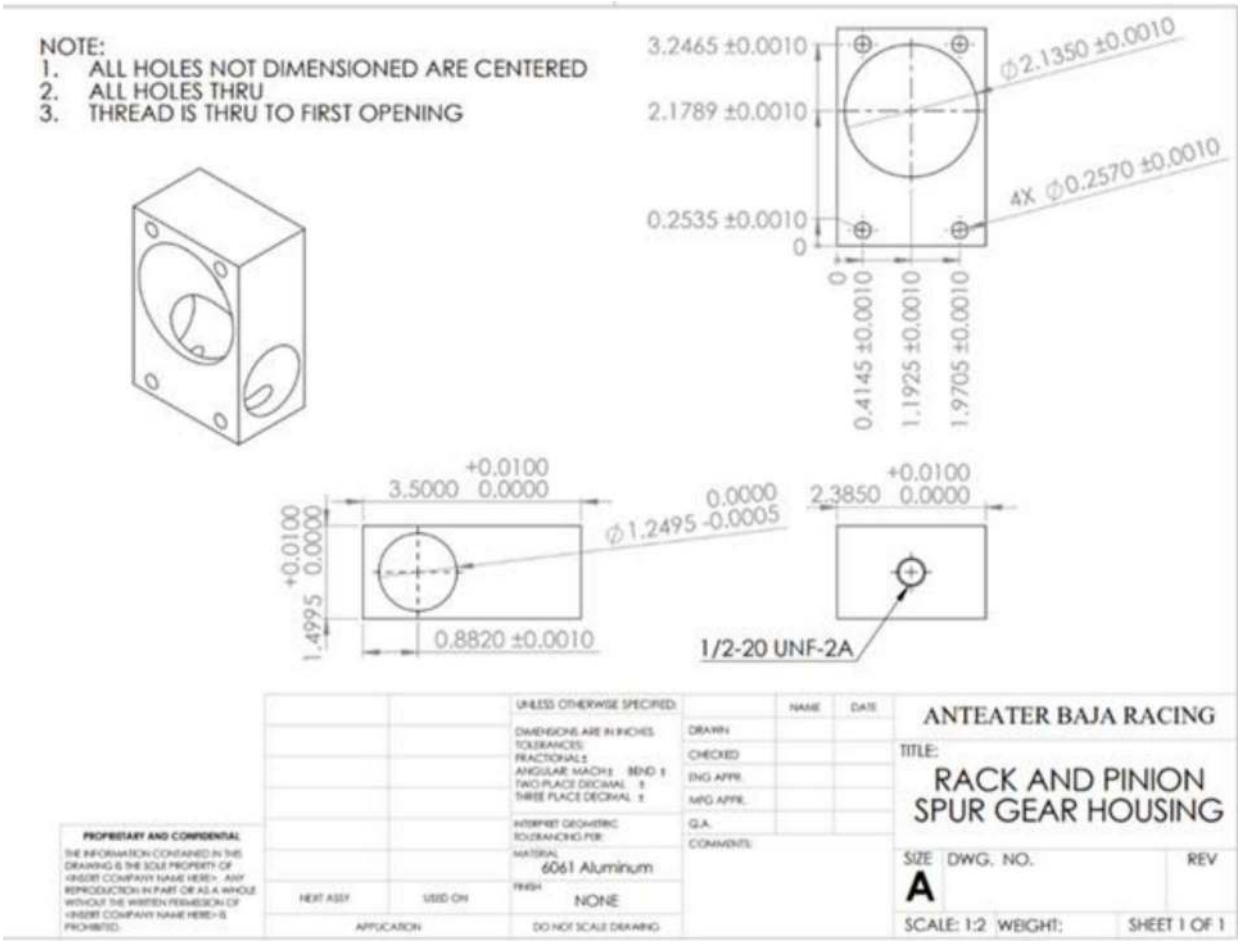


**Manufacturing**

Spur Gear Housing (Machined by Ellie Kirshner)

1. Cut 2.5” of stock material
2. Trim down stock to size (will need longer end mill for some sides, thickness is most important dimension)
3. Create 1.249-1.2495” hole for rack gear housing press fit (use program)
4. Press fit rack gear housing into spur gear housing (make sure to center rack housing)
5. Drill 4x 0.257” holes for bolts
6. Create 2.135” hole for spur gear (use program)
7. Drill and tap ½-20 UNF hole at bottom for bolt

Inner holes with program: use edge finder to find center, drill as big of a hole as you can in gradual increments, switch to collet and endmill, use spiral circle program (F4) to cut 25 thou below needed size, 0.015”/15% step over, full depth, switch to regular circle to get fully to size



**Manufacturing**

Housing Cap (Machined by Ellie Kirshner)

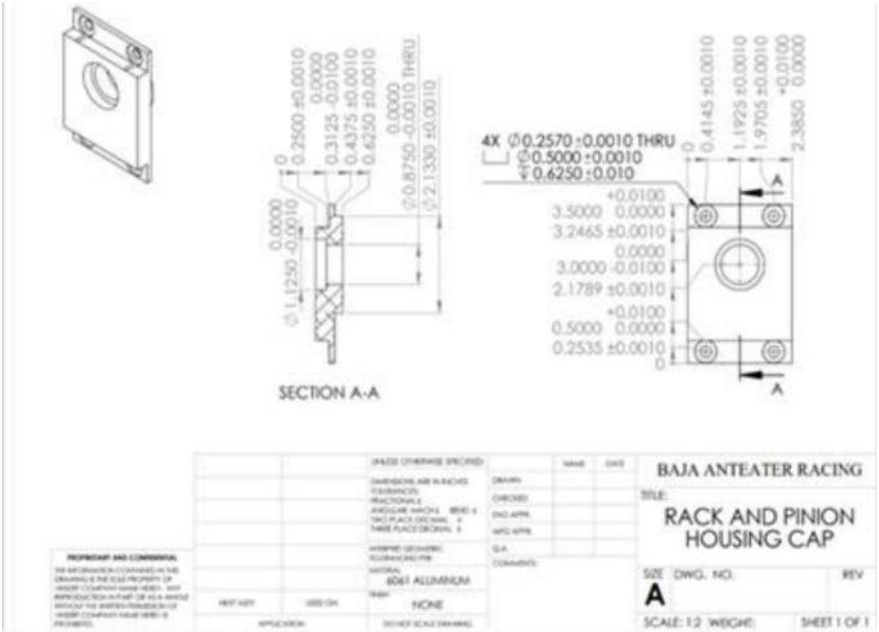
1. Cut 2.5” of stock material
2. Cut stock in half the long way (use vertical bandsaw)

Repeat for both pieces

1. Trim stock down to size (thickness is most important dimension)
2. Drill 4x 0.257” holes for bolts with countersink
3. Create 0.875” thru hole (use program)
4. Create 1.125” holes with 0.625” depth (use program)
5. Remove material from top and bottom of cap
6. Flip part and redo center
7. Create 2.133” outer hole (use program)
8. Remove extra surrounding material created by outer hole program

Inner holes with program: use edge finder to find center, drill as big of a hole as you can in gradual increments, switch to collet and endmill, use spiral circle program (F4) to cut 25 thou below needed size, 0.015”/15% step over, full depth, switch to regular circle to get fully to size

Outer holes with program: use edge finder to find center, use outer circle program (setting 0 for outer edge), 50 thou depth increments max



## Assembly

1. Press fit all seals and bearings
2. Cut and face ½”-20 bolt, cut ¼-28” bolt
3. Slide rack gear covered in red grease in and tighten bottom bolt (should not allow rotation, but too tight will increase steering effort significantly)
4. Fit spur gear into one side of housing cap
5. Cover spur gear in red grease and place into housing
6. Fit spur gear into other side of housing cap
7. Cut bellows to size and use zip tie to clamp onto housing
8. Add bolts and nuts to secure the caps and clevis (cap bolts need to be slightly loose to not clamp gear, clevis bolts need to be very tight to prevent rotation)
9. Clamp bellow to clevis with zip tie



## Integration

1. Set pillow block position by putting the bolts through bottom pillow block and the rack mount (6 washers on the diff side, no washers on the forward side)
   a. Pillow block has two orientations
2. Place rack and pinion onto pillow blocks (2nd furthest outboard position)
3. Remove bolts carefully, place top half of pillow block on, replace bolts
4. Secure with a washer and nut under the rack mount
5. Fit the steering column onto the spur gear spline
6. Attach the tie rods to the clevis



## Results Compared with Previous Years

1. 2026 - Custom Rack and Pinion: no failures during testing or competition, weight 4.4lbs, cost $194, 186 deg lock to lock w/ 3.25” of travel
2. 2025 - Kaz Rack and Pinion (OTS): very light (3lbs), very expensive ($850), hollow aluminum threaded rack extension sheared during 2025 competition due to lack of needed strength (repaired afterwards with 1144 steel threaded rack extension - new weight 5.3lbs), 248 deg lock to lock with 3.25” of travel
3. 2024 - EMPI Buggy Rack and Pinion (OTS): light (3.33lbs), cheap ($215), threaded rack extensions bent off during testing due to mounting too far inboard (fixed by adding long shaft collars for more support), 270 deg lock to lock w/ 3.875” of travel

## Maintenance

1. Every year should take apart all components, inspect, and reapply red grease